

第三章 向量

一、相关、无关

1、 n 维向量

n 个数 $a_1, a_2, \dots, a_n$ 构成的有序数组称为 n 维向量.

列向量: $\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ 或 $(a_1, a_2, \dots, a_n)^T$;

行向量: $(a_1, a_2, \dots, a_n)$.

零向量: 所有分量都是 0. [微信公众号: 考研斯基]

$$O = (0, 0, \dots, 0)^T$$

$$\text{设 } \alpha = (a_1, a_2, \dots, a_n)^T, \beta = (b_1, b_2, \dots, b_n)^T$$

$$\alpha = \beta \Leftrightarrow a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$$

运算

$$(1) \alpha + \beta = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)^T$$

$$(2) k\alpha = (ka_1, ka_2, \dots, ka_n)^T, \text{ 特别的 } 0\alpha = (0, 0, \dots, 0)^T = O$$

$$(3) (\alpha, \beta) = \alpha^T \beta = \beta^T \alpha = a_1 b_1 + a_2 b_2 + \dots + a_n b_n. \text{ 内积}$$

$$\text{正交 } (\alpha, \beta) = 0$$

$$\text{长度 } \alpha^T \alpha = a_1^2 + a_2^2 + \dots + a_n^2.$$

2、线性相关

对 n 维向量 $\alpha_1, \alpha_2, \dots, \alpha_s$, 如果存在不全为 0 的 $k_1, k_2, \dots, k_s$, 使得 $k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_s \alpha_s = 0$ 成立, 则称向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 否则称 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关;

【例 1】若 $\alpha_1 = (1, 3, 4, -2)^T, \alpha_2 = (2, 1, 3, t)^T, \alpha_3 = (3, -1, 2, 0)^T$ 线性相关, 则 $t = ?$

解: 设 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = 0$

$$\Rightarrow x_1 (1, 3, 4, -2)^T + x_2 (2, 1, 3, t)^T + x_3 (3, -1, 2, 0)^T = (0, 0, 0, 0)^T,$$

$$\text{即} \begin{cases} x_1 + 2x_2 + 3x_3 = 0, \\ 3x_1 + x_2 - x_3 = 0, \\ 4x_1 + 3x_2 + 2x_3 = 0, \\ -2x_1 + tx_2 = 0. \end{cases}$$

$$[\alpha_1, \alpha_2, \alpha_3] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & -1 \\ 4 & 3 & 2 \\ -2 & t & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -2t-2 \\ 0 & 0 & 0 \end{bmatrix}$$

$\alpha_1, \alpha_2, \alpha_3$ 线性相关 $\Leftrightarrow Ax = 0$ 有非 0 解 $\Leftrightarrow r(A) < n$, 故 $t = -1$.

【定理 1】向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关

$\Leftrightarrow \exists$ 不全为 0 的 $k_1, k_2, \dots, k_s$ 使 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$,

$$\Leftrightarrow \exists \text{ 不全为 0 的 } k_1, k_2, \dots, k_s \text{ 使 } [\alpha_1 \ \alpha_2 \ \dots \ \alpha_s] \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_s \end{bmatrix} = 0$$

$$\Leftrightarrow \text{齐次方程组 } [\alpha_1 \ \alpha_2 \ \dots \ \alpha_s] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_s \end{bmatrix} = 0 \text{ 有非 0 解}$$

$$\Leftrightarrow \text{秩 } r(\alpha_1 \ \alpha_2 \ \dots \ \alpha_s) < s$$

【推论】

(1) n 个 n 维向量 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性相关 $\Leftrightarrow |\alpha_1 \ \alpha_2 \ \dots \ \alpha_n| = 0$;

(2) $n+1$ 个 n 维向量必线性相关. [微信公众号: 考研斯基]

3、线性无关 (证 $\alpha_1 \ \alpha_2 \ \dots \ \alpha_s$ 无关)

$$\alpha_1 \ \alpha_2 \ \dots \ \alpha_s$$

$$0\alpha_1 + 0\alpha_2 + \dots + 0\alpha_s = 0$$

$$k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0 \quad ?$$

如果 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$ 必有 $k_1 = 0, k_2 = 0, \dots, k_s = 0$, 则称 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关.

当 $k_1, k_2, \dots, k_s$ 不全为 0 时, 必有 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s \neq 0$, 则称 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关.

(1) 定义法

$$\text{设 } k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$$



必有 $k_1 = 0, k_2 = 0, \dots, k_s = 0$.

【几何意义】

α 相关 $\Leftrightarrow \alpha = 0$;

α_1, α_2 相关 $\Leftrightarrow \alpha_1, \alpha_2$ 共线;

$\alpha_1, \alpha_2, \alpha_3$ 相关 $\Leftrightarrow \alpha_1, \alpha_2, \alpha_3$ 共面.

如 α_1, α_2 线性相关 $\Leftrightarrow \exists$ 不全为 0 的 k_1, k_2 使 $k_1\alpha_1 + k_2\alpha_2 = 0$.

不妨设 $k_1 \neq 0$, 有: $\alpha_1 = -\frac{k_2}{k_1}\alpha_2$,

如 $\alpha_1, \alpha_2, \alpha_3$ 相关 $\Leftrightarrow \exists$ 不全为 0 的 k_1, k_2, k_3 使 $k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 = 0$

对 $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$

$$k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 + 0\alpha_4 + 0\alpha_5 = 0$$

$k_1, k_2, k_3, 0, 0$ 不全为 0

如 $\alpha_1 \alpha_2 \dots \alpha_s$ 相关

【定理 2】一个向量小组, 如果有一个子集合相关, 它的整体一定相关. (向量个数增加)

如 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 则 $\alpha_1, \alpha_2, \dots, \alpha_s, \alpha_{s+1}, \dots, \alpha_t$ 必线性相关.

(如 $\exists$ 子集合相关 $\Rightarrow$ 整体必相关; 如向量组无关 $\Rightarrow$ 子集合必无关)

【定理 3】向量组线性无关, 它的延伸组也线性无关. (向量的坐标增加)

设 m 维向量 $\alpha_1, \alpha_2, \dots, \alpha_s$ 和 n 维向量 $\beta_1, \beta_2, \dots, \beta_s$ 构成 $m+n$ 维向量 $\gamma_1 = \begin{bmatrix} \alpha_1 \\ \beta_1 \end{bmatrix}, \gamma_2 = \begin{bmatrix} \alpha_2 \\ \beta_2 \end{bmatrix}, \dots,$

$\gamma_s = \begin{bmatrix} \alpha_s \\ \beta_s \end{bmatrix}$, 如 $\alpha_1, \alpha_2, \dots, \alpha_s$ 无关, 则 $\gamma_1, \gamma_2, \dots, \gamma_s$ 无关. [微信公众号: 考研斯基]

(低维无关 $\Rightarrow$ 延伸组高维必无关; 高维相关 $\Rightarrow$ 缩短组必相关)

示例: $\alpha_1, \alpha_2, \alpha_3$ 无关

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \text{ 只有 0 解}$$

$$\Rightarrow \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \text{ 只有 0 解}$$

即 $\begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix}, \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix}, \begin{pmatrix} \alpha_3 \\ \beta_3 \end{pmatrix}$ 无关

(2) 秩

证明 $r(\alpha_1 \alpha_2 \dots \alpha_s) = s$

➤ $r(A) = A$ 列秩 = A 行秩;

- $r(AB) \leq \min(r(A), r(B))$
- 如 A 可逆, $r(AB) = r(B), r(BA) = r(B)$;
- $A - m \times n$, 且 $r(A) = n$, 则 $r(AB) = r(B)$;
- $A - m \times n; B - n \times s$ 且 $AB = O \Rightarrow r(A) + r(B) \leq n$. [微信公众号: 考研斯基]

【例 2】已知 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 证明 $3\alpha_1 + 2\alpha_2, \alpha_2 - \alpha_3, 4\alpha_3 - 5\alpha_1$ 线性无关.

证明: 方法一 用定义

$$\text{设 } k_1(3\alpha_1 + 2\alpha_2) + k_2(\alpha_2 - \alpha_3) + k_3(4\alpha_3 - 5\alpha_1) = 0.$$

$$\text{即 } (3k_1 - 5k_3)\alpha_1 + (2k_1 + k_2)\alpha_2 + (-k_2 + 4k_3)\alpha_3 = 0.$$

$$\text{由 } \alpha_1, \alpha_2, \alpha_3 \text{ 无关, 得 } \begin{cases} 3k_1 - 5k_3 = 0 \\ 2k_1 + k_2 = 0 \\ -k_2 + 4k_3 = 0 \end{cases} \quad (1)$$

$$\text{因 } \begin{vmatrix} 3 & 0 & -5 \\ 2 & 1 & 0 \\ 0 & -1 & 4 \end{vmatrix} = 22 \neq 0, \text{ 齐次方程组 (1) 只有零解.}$$

$\therefore$ 必有 $k_1 = 0, k_2 = 0, k_3 = 0$, 从而 $3\alpha_1 + 2\alpha_2, \alpha_2 - \alpha_3, 4\alpha_3 - 5\alpha_1$ 必线性无关.

方法二 用秩

$$\text{令 } \beta_1 = 3\alpha_1 + 2\alpha_2, \beta_2 = \alpha_2 - \alpha_3, \beta_3 = 4\alpha_3 - 5\alpha_1$$

$$\text{则 } (\beta_1, \beta_2, \beta_3) = (3\alpha_1 + 2\alpha_2, \alpha_2 - \alpha_3, 4\alpha_3 - 5\alpha_1)$$

$$= (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} 3 & 0 & -5 \\ 2 & 1 & 0 \\ 0 & -1 & 4 \end{bmatrix}$$

$$\text{① 因 } \begin{bmatrix} 3 & 0 & -5 \\ 2 & 1 & 0 \\ 0 & -1 & 4 \end{bmatrix} \text{ 可逆, } \therefore r(\beta_1, \beta_2, \beta_3) = r(\alpha_1, \alpha_2, \alpha_3) = 3$$

从而 $\beta_1, \beta_2, \beta_3$ 线性无关, 即 $3\alpha_1 + 2\alpha_2, \alpha_2 - \alpha_3, 4\alpha_3 - 5\alpha_1$ 线性无关.

$$\text{② 因 } r(\alpha_1, \alpha_2, \alpha_3) = 3$$

$$\therefore r(\beta_1, \beta_2, \beta_3) = r \begin{bmatrix} 3 & 0 & -5 \\ 2 & 1 & 0 \\ 0 & -1 & 4 \end{bmatrix} = 3$$

即 $3\alpha_1 + 2\alpha_2, \alpha_2 - \alpha_3, 4\alpha_3 - 5\alpha_1$ 无关.

【例 3】 $A - n$ 阶, $\alpha - n$ 维列, $A^{m-1}\alpha \neq 0, A^m\alpha = 0$, 证明 $\alpha, A\alpha, A^2\alpha, \dots, A^{m-1}\alpha$ 无关.

证明: 设 $k_1\alpha + k_2A\alpha + k_3A^2\alpha + \dots + k_mA^{m-1}\alpha = 0 \quad (1)$

由 $A^m\alpha = 0 \Rightarrow A^{m+1}\alpha = 0, A^{m+2}\alpha = 0 \dots$

用 A^{m-1} 左乘 (1)

$$k_1 A^{m-1} \alpha = 0. \because A^{m-1} \alpha \neq 0, \therefore k_1 = 0$$

$$\text{代入 (1): } k_2 A \alpha + k_3 A^2 \alpha + \cdots + k_m A^{m-1} \alpha = 0 \quad (2).$$

类似地, 用 A^{m-2} 左乘 (2), 可得 $k_2 = 0$,

同理可知 $k_3 = 0, \cdots, k_m = 0$,

$\therefore \alpha, A\alpha, A^2\alpha, \cdots, A^{m-1}\alpha$ 线性无关.

【例 4】 $A - n$ 阶, $\alpha_1, \alpha_2, \alpha_3 - n$ 维列向量, $A\alpha_1 = \alpha_1 \neq 0, A\alpha_2 = \alpha_1 + \alpha_2, A\alpha_3 = \alpha_2 + \alpha_3$ 证明 $\alpha_1, \alpha_2, \alpha_3$ 线性无关.

证明: 定义法

$$\text{设 } k_1 \alpha_1 + k_2 \alpha_2 + k_3 \alpha_3 = 0 \quad (1)$$

$$\text{因为 } (A - E)\alpha_1 = 0, (A - E)\alpha_2 = \alpha_1, (A - E)\alpha_3 = \alpha_2$$

$$\text{用 } A - E \text{ 左乘 (1): } k_2 \alpha_1 + k_3 \alpha_2 = 0 \quad (2)$$

$$\text{用 } A - E \text{ 左乘 (2): } k_3 \alpha_1 = 0 \quad (3)$$

因 $\alpha_1 \neq 0$, 得 $k_3 = 0$

将 $k_3 = 0$ 代入 (2) 可得 $k_2 = 0, \dots$

故 $\alpha_1, \alpha_2, \alpha_3$ 线性无关.

【例 5】 $A - 3$ 阶, α_1, α_2 是 A 分别属于特征值 -1 和 1 的特征向量, 且 $A\alpha_3 = \alpha_2 + \alpha_3$, 证明 $\alpha_1, \alpha_2, \alpha_3$ 线性无关. [微信公众号: 考研斯基]

证明: 由特征向量定义, $A\alpha_1 = -\alpha_1, A\alpha_2 = \alpha_2$.

$$\text{设 } k_1 \alpha_1 + k_2 \alpha_2 + k_3 \alpha_3 = 0 \quad (1)$$

$$\text{用 } A \text{ 左乘 (1): } -k_1 \alpha_1 + k_2 \alpha_2 + k_3 (\alpha_2 + \alpha_3) = 0 \quad (2)$$

$$(1) - (2): 2k_1 \alpha_1 - k_3 \alpha_2 = 0 \quad (3)$$

因 α_1, α_2 是不同特征值的特征向量, 有 α_1, α_2 必线性无关.

由 (3)

$$\begin{cases} 2k_1 = 0 \\ -k_3 = 0 \end{cases}, \therefore k_1 = 0, k_3 = 0$$

代入 (1) 得 $k_2 \alpha_2 = 0$, 因 α_2 是特征向量, 知 $\alpha_2 \neq 0$

从而 $k_2 = 0$, 故 $\alpha_1, \alpha_2, \alpha_3$ 必线性无关.

【例 6】 设 λ_1, λ_2 是矩阵 A 不同的特征值, α_1, α_2 是特征值 λ_1 线性无关的特征向量, α 是特征值 λ_2 的特征向量. 证明: $\alpha_1, \alpha_2, \alpha$ 线性无关.

证明: 按已知有 $A\alpha_1 = \lambda_1 \alpha_1, A\alpha_2 = \lambda_1 \alpha_2, A\alpha = \lambda_2 \alpha$.

$$\text{如 } k_1\alpha_1 + k_2\alpha_2 + k\alpha = 0 \quad (1)$$

$$\text{用 } A \text{ 左乘 (1): } k_1\lambda_1\alpha_1 + k_2\lambda_1\alpha_2 + k\lambda_2\alpha = 0 \quad (2)$$

$$\text{用 } \lambda_1 \text{ 左乘 (1): } \lambda_1 k_1\alpha_1 + \lambda_1 k_2\alpha_2 + \lambda_1 k\alpha = 0 \quad (3)$$

$$(2) - (3): k(\lambda_2 - \lambda_1)\alpha = 0, \text{ 因 } \lambda_1 \neq \lambda_2, \alpha \neq 0$$

$$\therefore k = 0$$

$$\text{代入 (1): } k_1\alpha_1 + k_2\alpha_2 = 0$$

由 α_1, α_2 线性无关

$$\therefore k_1 = 0, k_2 = 0$$

故 $\alpha_1, \alpha_2, \alpha$ 线性无关. [微信公众号: 考研斯基]

【定理证明】如 $\beta_1, \beta_2, \beta_3$ 可由 α_1, α_2 线性表示, 则 $\beta_1, \beta_2, \beta_3$ 线性相关.

$$\text{证明: 设 } \beta_1 = C_{11}\alpha_1 + C_{21}\alpha_2$$

$$\beta_2 = C_{12}\alpha_1 + C_{22}\alpha_2$$

$$\beta_3 = C_{13}\alpha_1 + C_{23}\alpha_2$$

$$\text{即 } (\beta_1 \beta_2 \beta_3) = (\alpha_1 \alpha_2) \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \end{bmatrix}$$

$$\begin{aligned} \text{则 } r(\beta_1 \beta_2 \beta_3) &= r[(\alpha_1 \alpha_2) C] \\ &\leq r(\alpha_1 \alpha_2) \leq 2 < 3 \end{aligned}$$

【例 7】 $A - m \times n, r(A) = n, \alpha_1, \alpha_2, \alpha_3, n$ 维无关. 证明 $A\alpha_1, A\alpha_2, A\alpha_3$ 线性无关.

证明: 方法一 用定义

$$\text{设 } k_1A\alpha_1 + k_2A\alpha_2 + k_3A\alpha_3 = 0 \quad (1)$$

$$\text{即 } A(k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3) = 0$$

$$\text{因 } A - m \times n, r(A) = n,$$

齐次方程组 $Ax = 0$ 只有 0 解,

$$\text{故 } k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 = 0 \quad (2)$$

$$\text{由 } \alpha_1, \alpha_2, \alpha_3 \text{ 线性无关, 从而 } k_1 = 0, k_2 = 0, k_3 = 0$$

所以 $A\alpha_1, A\alpha_2, A\alpha_3$ 线性无关.

方法二 用秩

$$(A\alpha_1, A\alpha_2, A\alpha_3) = A(\alpha_1, \alpha_2, \alpha_3)$$

$$\text{如 } A - m \times n, r(A) = n, \text{ 则 } r(AB) = r(B)$$

$$r(A\alpha_1, A\alpha_2, A\alpha_3) = r(\alpha_1, \alpha_2, \alpha_3) = 3.$$

$\therefore A\alpha_1, A\alpha_2, A\alpha_3$ 线性无关.

【例 8】设 4 维列向量 $\alpha_1, \alpha_2, \alpha_3$ 无关, 且与非 0 的 β_1, β_2 均正交, 证明

(1) β_1, β_2 相关;

(2) $\alpha_1, \alpha_2, \alpha_3, \beta_1$ 无关.

证明: (1) 构造 $A = \begin{bmatrix} \alpha_1^T \\ \alpha_2^T \\ \alpha_3^T \end{bmatrix} - 3 \times 4$

由 $\alpha_1, \alpha_2, \alpha_3$ 无关, 知 $r(A)=3$

因 α_i 与 β_j 正交 $i = 1, 2, 3 \quad j = 1, 2$

$$A\beta_j = \begin{bmatrix} \alpha_1^T \\ \alpha_2^T \\ \alpha_3^T \end{bmatrix} \beta_j = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

β_1, β_2 均是齐次方程组 $Ax = 0$ 的解, 于是

$$r(\beta_1, \beta_2) \leq n - r(A) = 4 - 3 = 1,$$

$\therefore \beta_1, \beta_2$ 线性相关.

(2) 定义法 设 $k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 + k\beta_1 = 0$ (1)

因 α_i 与 β_1 正交, 内积 $\beta_1^T \alpha_i = 0 (i = 1, 2, 3)$

用 β_1^T 左乘 (1)

$$k_1\beta_1^T\alpha_1 + k_2\beta_1^T\alpha_2 + k_3\beta_1^T\alpha_3 + k\beta_1^T\beta_1 = 0$$

$$\text{有 } k\beta_1^T\beta_1 = 0. \quad (2)$$

$$\text{由 } \beta_1 \neq 0, \text{ 知 } \beta_1^T\beta_1 \neq 0 \quad \therefore k = 0.$$

把 $k = 0$ 代入 (1)

$$k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 = 0,$$

因 $\alpha_1, \alpha_2, \alpha_3$ 线性无关

$$\therefore k_1 = 0, k_2 = 0, k_3 = 0,$$

所以 $\alpha_1, \alpha_2, \alpha_3, \beta_1$ 线性无关. [微信公众号: 考研斯基]

【例 9】已知 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, $\alpha_1 + a\alpha_2, \alpha_1 + 2\alpha_2 + \alpha_3, a\alpha_1 - \alpha_3$ 线性相关, 则 $a = \underline{\hspace{1cm}}$

解: 用秩

$$\beta_1 = \alpha_1 + a\alpha_2, \beta_2 = \alpha_1 + 2\alpha_2 + \alpha_3, \beta_3 = a\alpha_1 - \alpha_3$$

$$\text{则 } (\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} 1 & 1 & a \\ a & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\beta_1, \beta_2, \beta_3 \text{ 相关} \Leftrightarrow r(\beta_1, \beta_2, \beta_3) < 3$$

$$\text{由 } \alpha_1, \alpha_2, \alpha_3 \text{ 无关} \Leftrightarrow r(\alpha_1, \alpha_2, \alpha_3) = 3$$

$A = (\alpha_1, \alpha_2, \alpha_3)$ 列满秩

$$\therefore \beta_1, \beta_2, \beta_3 \text{ 相关} \Leftrightarrow \begin{vmatrix} 1 & 1 & a \\ a & 2 & 0 \\ 0 & 1 & -1 \end{vmatrix} = a^2 + a - 2 = 0$$

故 $a = 1$ 或 $a = -2$.

【例 10】已知 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 下列向量组中线性无关的是 ().

(A) $\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1$

(B) $\alpha_1 + \alpha_2, \alpha_2 + 2\alpha_3, \alpha_1 + 2\alpha_2 + \alpha_3, \alpha_1 - \alpha_2 + 5\alpha_3$

(C) $\alpha_1 + 2\alpha_2, 2\alpha_2 + 3\alpha_3, 3\alpha_3 - \alpha_1$

(D) $\alpha_1 + \alpha_2 - \alpha_3, 2\alpha_1 + 3\alpha_2 + 12\alpha_3, 3\alpha_1 + 5\alpha_2 + 25\alpha_3$

解: 排除 (B), 4 个向量可由 3 个向量线性表出, (B) 线性相关.

排除 (C), $(\alpha_1 + 2\alpha_2) - (2\alpha_2 + 3\alpha_3) + (3\alpha_3 - \alpha_1) = 0$, (C) 相关

排除 (A), $(\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, 因 $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} \neq 0$, 故

$r(\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) = r(\alpha_1, \alpha_2, \alpha_3) = 3$, 所以 (A) 线性相关.

二、线性表示

1、线性组合

设 $\alpha_1, \alpha_2, \dots, \alpha_s$ 是 n 维向量, $k_1, k_2, \dots, k_s$ 是一组实数, 称 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s$ 是向量 $\alpha_1, \alpha_2, \dots, \alpha_s$ 的线性组合. [微信公众号: 考研斯基]

2、线性表示

对 n 维向量 $\alpha_1, \alpha_2, \dots, \alpha_s$ 和 β , 若 $\exists$ 实数 $k_1, k_2, \dots, k_s$ 使得 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = \beta$ 成立, 则称向量 β 是向量 $\alpha_1, \alpha_2, \dots, \alpha_s$ 的线性组合, 或称向量 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性表示.

【定理 4】向量 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性表出 $\Leftrightarrow \exists$ 实数 $k_1, k_2, \dots, k_s$ 使 $k_1\alpha_1 + k_2\alpha_2 + \dots +$

$$k_s\alpha_s = \beta \Leftrightarrow \exists \text{ 实数 } k_1, k_2, \dots, k_s \text{ 使 } (\alpha_1 \alpha_2 \cdots \alpha_s) \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_s \end{bmatrix} = \beta \Leftrightarrow \text{方程组 } (\alpha_1 \alpha_2 \cdots \alpha_s) \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_s \end{bmatrix} = \beta \text{ 有}$$

$$\text{解} \Leftrightarrow \text{秩} r(\alpha_1 \alpha_2 \cdots \alpha_s) = r(\alpha_1 \alpha_2 \cdots \alpha_s \beta)$$

【例 11】 $\alpha_1 = (1, 2, 3, a)^T, \alpha_2 = (1, 1, 2, -a)^T, \alpha_3 = (3, 5, b+4, 2)^T, \beta = (3, 4, 7, -2)^T$. 试讨论当 a, b 为何值时,

(1) β 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示;

(2) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 并求出表示式. [微信公众号: 考研斯基]

解: 设 $x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta$, 对 $[\alpha_1, \alpha_2, \alpha_3, \beta]$ 作初等行变换:

$$\bar{A} = \left[\begin{array}{cccc|c} 1 & 1 & 3 & 3 & 3 \\ 2 & 1 & 5 & 4 & 4 \\ 3 & 2 & b+4 & 7 & 7 \\ a & -a & 2 & -2 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 1 \\ & 1 & 1 & 1 & 2 \\ & & b-4 & 0 & 0 \\ & & 2-a & a-2 & a-2 \end{array} \right]$$

如 $a \neq 2$

$$\bar{A} \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 1 \\ & 1 & 1 & 2 & 2 \\ & & 1 & -1 & -1 \\ & & 0 & b-4 & b-4 \end{array} \right] \xrightarrow{b=4} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 3 \\ & 1 & 0 & 3 & 3 \\ & & 1 & -1 & -1 \\ & & 0 & 0 & 0 \end{array} \right]$$

(1) $a \neq 2$, 且 $b \neq 4$: 则 $r(A) = 3, r(\bar{A}) = 4, r(A) \neq r(\bar{A}), \beta$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

(2) $a \neq 2, b = 4$: 则 $x_1 = 3, x_2 = 3, x_3 = -1, \beta = 3\alpha_1 + 3\alpha_2 - \alpha_3$.

如 $a = 2$

$$\bar{A} \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 1 \\ & 1 & 1 & 2 & 2 \\ & & b-4 & 0 & 0 \\ & & 0 & 0 & 0 \end{array} \right]$$

$$(3) a = 2, b \neq 4: \bar{A} \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 1 \\ & 1 & 0 & 2 & 2 \\ & & 1 & 0 & 0 \\ & & 0 & 0 & 0 \end{array} \right], r(A) = r(\bar{A}) = n, \text{ 解得 } x_1 = 1, x_2 = 2, x_3 = 0,$$

所以 $\beta = \alpha_1 + 2\alpha_2$.

$$(4) a = 2, b = 4: \bar{A} \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 1 \\ & 1 & 1 & 2 & 2 \\ & & 0 & 0 & 0 \\ & & & 0 & 0 \end{array} \right], r(A) = r(\bar{A}) = 2 < 3,$$

令 $x_3 = t \Rightarrow x_1 = 1 - 2t, x_2 = 2 - t, \beta = (1 - 2t)\alpha_1 + (2 - t)\alpha_2 + t\alpha_3. \forall t$

3、向量组的线性表出

向量组 (I) $\alpha_1 \alpha_2 \cdots \alpha_s$; (II) $\beta_1 \beta_2 \cdots \beta_t$, 如 $\forall \alpha_i, \alpha_i$ 均可由 $\beta_1 \beta_2 \cdots \beta_t$ 线性表出, 则称向量组 (I) 可由 (II) 线性表出.

如 $\exists \alpha_i, \alpha_i$ 不能由 $\beta_1 \beta_2 \cdots \beta_t$ 线性表出, 则称向量组 (I) 不能由向量组 (II) 线性表出.

【例 12】(I) $\alpha_1 = [1, 1, a]^T, \alpha_2 = [1, a, 1]^T, \alpha_3 = [a, 1, 1]^T$, (II) $\beta_1 = [1, 1, a]^T, \beta_2 = [-2, a, 4]^T, \beta_3 = [-2, a, a]^T$, (I) 可由 (II) 线性表示, 但 (II) 不能由 (I) 线性表示, 求 a .

解: (I) 可由 (II) 线性表示, 即方程组

$$x_{1j}\beta_1 + x_{2j}\beta_2 + x_{3j}\beta_3 = \alpha_j (j = 1, 2, 3)$$

同时有解:

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -2 & 1 & 1 & a \\ 1 & a & a & 1 & a & 1 \\ a & 4 & a & a & 1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & -2 & 1 & 1 & a \\ 0 & a+2 & a+2 & 0 & a-1 & 1-a \\ 0 & 0 & a-4 & 0 & 3-3a & -(a-1)^2 \end{array} \right]$$

$\forall a, \alpha_1$ 可由 (II) 线性表示

$a \neq 4$ 且 $a \neq -2, \alpha_2$ 可由 (II) 线性表示;

$a \neq 4$ 且 $a \neq -2, \alpha_3$ 可由 (II) 线性表示

$\therefore a \neq 4$ 且 $a \neq -2$ 时 (I) 可由 (II) 线性表示.

(II) 不能由 (I) 线性表示, 即

$$x_{1j}\alpha_1 + x_{2j}\alpha_2 + x_{3j}\alpha_3 = \beta_j (j = 1, 2, 3)$$

至少有一个方程组无解

$$\left[\begin{array}{ccc|ccc} 1 & 1 & a & 1 & -2 & -2 \\ 1 & a & 1 & 1 & a & a \\ a & 1 & 1 & a & 4 & a \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & a & 1 & -2 & -2 \\ 0 & a-1 & 1-a & 0 & a+2 & a+2 \\ 0 & 0 & 2-a-a^2 & 0 & 3a+6 & 4a+2 \end{array} \right]$$

$\forall a, \beta_1$ 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

$a = 1$ 时, β_2 不能由 (I) 线性表示;

$a = 1$ 或 $a = -2$ 时, β_3 不能由 (I) 线性表示

$\therefore a = 1$ 或 $a = -2$ 时, (II) 不能由 (I) 线性表示.

故 $a = 1$ 时 (I) 可由 (II) 线性表出, 而 (II) 不能由 (I) 线性表示.

【定理 5】 $\alpha_1, \alpha_2, \cdots, \alpha_s$ 相关 ($s \geq 2$) $\Leftrightarrow \exists$ 某 α_i 可由 $\alpha_1, \cdots, \alpha_{i-1}, \alpha_{i+1}, \cdots, \alpha_s$ 表出

【定理 6】如 $\alpha_1, \alpha_2, \cdots, \alpha_s$ 无关, $\alpha_1, \alpha_2, \cdots, \alpha_s, \beta$ 相关, 则 β 可由 $\alpha_1, \alpha_2, \cdots, \alpha_s$ 线性表出, 且表示法唯一. [微信公众号: 考研斯基]

【定理 7】如 $\alpha_1, \alpha_2, \dots, \alpha_s$ 可由 $\beta_1, \beta_2, \dots, \beta_t$ 表出, 且 $s > t$, 则 $\alpha_1, \alpha_2, \dots, \alpha_s$ 必相关 (多由少表出, 则多必相关).

$$\text{如 } \beta_1 = C_{11}\alpha_1 + C_{21}\alpha_2 + C_{31}\alpha_3$$

$$\beta_2 = C_{12}\alpha_1 + C_{22}\alpha_2 + C_{32}\alpha_3$$

$$\beta_3 = C_{13}\alpha_1 + C_{23}\alpha_2 + C_{33}\alpha_3$$

$$\beta_4 = C_{14}\alpha_1 + C_{24}\alpha_2 + C_{34}\alpha_3$$

$$(\beta_1, \beta_2, \beta_3, \beta_4) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} \\ C_{21} & C_{22} & C_{23} & C_{24} \\ C_{31} & C_{32} & C_{33} & C_{34} \end{bmatrix}$$

$$r(\beta_1, \beta_2, \beta_3, \beta_4) = r[(\alpha_1, \alpha_2, \alpha_3)C]$$

$$\leq r(\alpha_1, \alpha_2, \alpha_3) \leq 3 < 4$$

$\therefore \beta_1, \beta_2, \beta_3, \beta_4$ 必线性相关. [微信公众号: 考研斯基]

【推论】如 $\alpha_1, \alpha_2, \dots, \alpha_s$ 无关, 且 $\alpha_1, \alpha_2, \dots, \alpha_s$ 可由 $\beta_1, \beta_2, \dots, \beta_t$ 线性表出, 则 $s \leq t$.

【定理 8】如 $\alpha_1, \alpha_2, \dots, \alpha_s$ 可由 $\beta_1, \beta_2, \dots, \beta_t$ 线性表出, 则 $r(\alpha_1, \alpha_2, \dots, \alpha_s) \leq r(\beta_1, \beta_2, \dots, \beta_t)$.

【推论】如向量组 (I) 和 (II) 等价, 则 $r(I) = r(II)$.

【例 13】 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, $\alpha_2, \alpha_3, \alpha_4$ 无关.

(I) α_1 能否由 α_2, α_3 线性表出? 给出你的证明.

(II) α_4 能否由 $\alpha_1, \alpha_2, \alpha_3$ 线性表出? 给出你的证明.

解: (I) 方法一 $\because \alpha_2, \alpha_3, \alpha_4$ 无关 $\therefore \alpha_2, \alpha_3$ 线性无关.

又因 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 故 α_1 必可由 α_2, α_3 线性表示.

方法二 $\because \alpha_1, \alpha_2, \alpha_3$ 相关, 故 $\exists$ 不全为 0 的 k_1, k_2, k_3 , 使

$$k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 = 0 \quad (1)$$

如 $k_1 = 0$, 则 k_2, k_3 不全为 0, 而

$$k_2\alpha_2 + k_3\alpha_3 = 0 \quad (2)$$

于是 α_2, α_3 线性相关

那么 $\alpha_2, \alpha_3, \alpha_4$ 线性相关, 与已知矛盾

故必有 $k_1 \neq 0$, 由 (1) 得

$$\alpha_1 = -\frac{k_2}{k_1}\alpha_2 - \frac{k_3}{k_1}\alpha_3. \text{ 即 } \alpha_1 \text{ 可由 } \alpha_2, \alpha_3 \text{ 线性表出}$$

(II) 方法一 反证法

如 α_4 能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 设

$$\alpha_4 = k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3$$

由 (I) 有 $\alpha_1 = l_2\alpha_2 + l_3\alpha_3$, 则 $\alpha_4 = k_1(l_2\alpha_2 + l_3\alpha_3) + k_2\alpha_2 + k_3\alpha_3$

$$= (k_1l_2 + k_2)\alpha_2 + (k_1l_3 + k_3)\alpha_3$$

于是 $\alpha_2, \alpha_3, \alpha_4$ 线性相关, 与已知矛盾.

$\therefore \alpha_4$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

方法二 秩

对于方程组 $x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \alpha_4$

由 $\alpha_1, \alpha_2, \alpha_3$ 相关, 知 $r(\alpha_1, \alpha_2, \alpha_3) < 3$

又因 $\alpha_2, \alpha_3, \alpha_4$ 线性无关, 知 $r(\alpha_1, \alpha_2, \alpha_3, \alpha_4) \geq r(\alpha_2, \alpha_3, \alpha_4) = 3$.

即 $r(A) \neq r(\bar{A})$, 方程组无解

$\therefore \alpha_4$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表出.

【例 14】 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性表出, 但 β 不能由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 线性表出.

(I) α_m 能否由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}, \beta$ 线性表出?

(II) α_m 能否由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 线性表出?

解: (I) β 可由 $\alpha_1, \alpha_2, \dots, \alpha_m$ 表出 $\Leftrightarrow \exists l_1, l_2, \dots, l_m$, 使

$$\beta = l_1\alpha_1 + l_2\alpha_2 + \dots + l_{m-1}\alpha_{m-1} + l_m\alpha_m. \quad (1)$$

由 β 不能由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 线性表出

$$\text{必有 } l_m \neq 0, \text{ 从而 } \alpha_m = \frac{1}{l_m}(\beta - l_1\alpha_1 - l_2\alpha_2 - \dots - l_{m-1}\alpha_{m-1}) \quad (2)$$

故 α_m 必可由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}, \beta$ 线性表出.

(II) 反证法

如 α_m 能由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 线性表出

$$\text{设 } \alpha_m = k_1\alpha_1 + k_2\alpha_2 + \dots + k_{m-1}\alpha_{m-1}$$

$$\text{那么 } \beta = l_1\alpha_1 + l_2\alpha_2 + \dots + l_{m-1}\alpha_{m-1} + l_m(k_1\alpha_1 + k_2\alpha_2 + \dots + k_{m-1}\alpha_{m-1})$$

$$= (l_1 + l_mk_1)\alpha_1 + (l_2 + l_mk_2)\alpha_2 + \dots + (l_{m-1} + l_mk_{m-1})\alpha_{m-1}$$

与 β 不能由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 线性表出相矛盾

故 α_m 不能由 $\alpha_1, \alpha_2, \dots, \alpha_{m-1}$ 表出. [微信公众号: 考研斯基]

【注】 线性表出题目常用方法:

(1) 构造方程组, 证明方程组有解

$$r(\alpha_1, \alpha_2, \dots, \alpha_s) = r(\alpha_1, \alpha_2, \dots, \alpha_s, \beta).$$

(2) 找出两个条件: $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性无关, $\alpha_1, \alpha_2, \dots, \alpha_s, \beta$ 线性相关.

(3) 证 $k \neq 0$ (能表出).

(4) 反证法 (不能表出) .

【例 15】设 $A = [\alpha_1, \alpha_2, \dots, \alpha_m], B = [\beta_1, \beta_2, \dots, \beta_m], P$ 可逆, 且 $PA = B$.

(1) 如 $\alpha_1, \alpha_3, \alpha_5$ 线性无关, 则 $\beta_1, \beta_3, \beta_5$ 无关.

(2) 如 $\alpha_m = 2\alpha_1 + 5\alpha_3$, 则 $\beta_m = 2\beta_1 + 5\beta_3$.

证: $PA = B, P[\alpha_1, \alpha_2, \dots, \alpha_m] = [\beta_1, \beta_2, \dots, \beta_m]$

$$P\alpha_i = \beta_i (i = 1, 2, \dots, m)$$

(1) 方法一

$$\text{设 } k_1\beta_1 + k_3\beta_3 + k_5\beta_5 = 0$$

$$\text{即 } k_1P\alpha_1 + k_3P\alpha_3 + k_5P\alpha_5 = 0$$

因 P 可逆, 左乘 P^{-1} , 得

$$k_1\alpha_1 + k_3\alpha_3 + k_5\alpha_5 = 0$$

由 $\alpha_1, \alpha_3, \alpha_5$ 线性无关

$$\therefore k_1 = 0, k_3 = 0, k_5 = 0$$

故 $\beta_1, \beta_3, \beta_5$ 必无关.

方法二 $r(\beta_1, \beta_3, \beta_5) = r[P(\alpha_1, \alpha_3, \alpha_5)] = r(\alpha_1, \alpha_3, \alpha_5)$

(2) 由 $\alpha_m = 2\alpha_1 + 5\alpha_3$, 则 $P\alpha_m = P(2\alpha_1 + 5\alpha_3) = 2P\alpha_1 + 5P\alpha_3$

$$\text{即 } \beta_m = 2\beta_1 + 5\beta_3.$$

三、向量组的秩

1、极大线性无关组

在向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 中, 若存在 $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_r}$ 线性无关, 而再添加任一个 $\alpha_j (j = 1, 2, \dots, s)$ 就有 $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_r}, \alpha_j$ 线性相关, 则称 $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_r}$ 是向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 的一个极大线性无关组. [微信公众号: 考研斯基]

2、向量组的秩

向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 的极大线性无关组中所含向量的个数 r 称为向量组的秩, 记为 $r(\alpha_1, \alpha_2, \dots, \alpha_s) = r$.

【例 16】 $r(\alpha_1, \alpha_2, \dots, \alpha_s) = s$

$\Leftrightarrow$ 极大线性无关组中有 s 个向量

$\Leftrightarrow$ 极大: $\alpha_1, \alpha_2, \dots, \alpha_s$

$\Leftrightarrow \alpha_1, \alpha_2, \dots, \alpha_s$ 无关

$\alpha_1, \alpha_2, \dots, \alpha_s$ 无关 $(\forall k_i \equiv 0)$

$$\Leftrightarrow [\alpha_1, \alpha_2, \dots, \alpha_s] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_s \end{bmatrix} = 0 \text{ 只有 } 0 \text{ 解}$$

$$\Leftrightarrow r(\alpha_1, \alpha_2, \dots, \alpha_s) = s.$$

【例 17】 $\alpha_1 = (1, 1, 4, 2)^T, \alpha_2 = (1, -1, -2, 4)^T, \alpha_3 = (-3, 1, a, -10)^T, \alpha_4 = (1, 3, 10, 0)^T$, 求向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 的秩和极大无关组, 并将其他向量用极大无关组线性表示.

解: 对 $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ 作初等行变换

$$[\alpha_1, \alpha_2, \alpha_3, \alpha_4] = \begin{bmatrix} 1 & 1 & -3 & 1 \\ 1 & -1 & 1 & 3 \\ 4 & -2 & a & 10 \\ 2 & 4 & -10 & 0 \end{bmatrix} \xrightarrow{\text{行}} \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(1) 如 $a \neq 0$

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{秩 } r(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = 3$$

极大无关组: $\alpha_1, \alpha_2, \alpha_3$

$$\alpha_4 = 2\alpha_1 - \alpha_2.$$

(2) $a = 0$

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) \rightarrow \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow r(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = 2$$

极大无关组: α_1, α_2

$$\alpha_3 = -\alpha_1 - 2\alpha_2; \quad \alpha_4 = 2\alpha_1 - \alpha_2. \quad [\text{微信公众号: 考研斯基}]$$

【补充】

当 $a \neq 0$ 时

$$\beta_4 = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = 2\beta_1 - \beta_2, \text{ 于是 } \alpha_4 = 2\alpha_1 - \alpha_2.$$

【例 18】 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 若

$$\beta_1 = (a+1)\alpha_1 + \alpha_2 + \alpha_3,$$

$$\beta_2 = 3\alpha_1 + (a+3)\alpha_2 + 3\alpha_3,$$

$$\beta_3 = 5\alpha_1 + 5\alpha_2 + (a+5)\alpha_3.$$

若 $\beta_1, \beta_2, \beta_3$ 线性相关, 求 $r(\beta_1, \beta_2, \beta_3)$, 极大无关组, 线性表出.

$$\text{解: } (\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} a+1 & 3 & 5 \\ 1 & a+3 & 5 \\ 1 & 3 & a+5 \end{bmatrix}.$$

$$\text{由 } r(\alpha_1, \alpha_2, \alpha_3) = 3 \Rightarrow r(\beta_1, \beta_2, \beta_3) = r(A)$$

$$\beta_1, \beta_2, \beta_3 \text{ 相关} \Leftrightarrow \begin{vmatrix} a+1 & 3 & 5 \\ 1 & a+3 & 5 \\ 1 & 3 & a+5 \end{vmatrix} = 0$$

$$|A| = a^2(a+9)$$

$$(1) a=0 \text{ 时}, r(\beta_1, \beta_2, \beta_3) = 1$$

$$\text{极大: } \beta_1 = \alpha_1 + \alpha_2 + \alpha_3, \beta_2 = 3\beta_1, \beta_3 = 5\beta_1.$$

$$(2) \text{ 如 } a=-9, A = \begin{bmatrix} -8 & 3 & 5 \\ 1 & -6 & 5 \\ 1 & 3 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} -8 & 3 & 5 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

$$r(\beta_1, \beta_2, \beta_3) = 2$$

$$\text{极大: } \beta_2, \beta_3, \beta_1 = -\beta_2 - \beta_3.$$

3、线性表示与秩

【定理 9】如向量组 (I) 可由向量组 (II) 线性表出, 则 $r(I) \leq r(II)$.

向量组等价:

(1) 如果向量组 (I) 和 (II) 可以互相线性表出, 则称向量组等价.

(2) 向量组和其极大线性无关组等价.

(3) 向量组的两个极大线性无关组等价.

(4) 如向量组 (I) 和 (II) 等价, 则 $r(I) = r(II)$.

(5) 如 $r(I) = r(II)$, 且 (I) 可由 (II) 线性表出 $\Rightarrow$ (I) 与 (II) 等价.

(6) 如 $r(I) = r(II) = r(I, II) \Rightarrow$ (I) 与 (II) 等价. [微信公众号: 考研斯基]

【例 12】确定常数 a , 使向量组 $\alpha_1 = (1, 1, a)^T, \alpha_2 = (1, a, 1)^T, \alpha_3 = (a, 1, 1)^T$ 可由向量组 $\beta_1 = (1, 1, a)^T, \beta_2 = (-2, a, 4)^T, \beta_3 = (-2, a, a)^T$ 线性表示, 但向量组 $\beta_1, \beta_2, \beta_3$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

解: $\because \alpha_1, \alpha_2, \alpha_3$ 可由 $\beta_1, \beta_2, \beta_3$ 线性表出

$$\therefore r(\alpha_1, \alpha_2, \alpha_3) \leq r(\beta_1, \beta_2, \beta_3)$$

又因 $\beta_1, \beta_2, \beta_3$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 表出

$$\therefore r(\alpha_1, \alpha_2, \alpha_3) < r(\beta_1, \beta_2, \beta_3) \leq 3.$$

$$\Rightarrow r(\alpha_1, \alpha_2, \alpha_3) < 3$$

$$\Rightarrow |\alpha_1, \alpha_2, \alpha_3| = \begin{vmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{vmatrix} = -(a+2)(a-1)^2 = 0$$

$$\therefore a = 1 \text{ 或 } a = -2$$

当 $a = 1$ 时, $\alpha_1 = \alpha_2 = \alpha_3 = \beta_1$, 即 $\alpha_1, \alpha_2, \alpha_3$ 可由 $\beta_1, \beta_2, \beta_3$ 线性表出, 而 $\beta_2 = (-2, 1, 4)^T$ 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表出.

$$\text{当 } a = -2 \text{ 时, } [\alpha_1, \alpha_2, \alpha_3] = \begin{bmatrix} 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{bmatrix}$$

$$[\beta_1, \beta_2, \beta_3] = \begin{bmatrix} 1 & -2 & -2 \\ 1 & -2 & -2 \\ -2 & 4 & -2 \end{bmatrix}$$

$$\text{有 } r(\alpha_1, \alpha_2, \alpha_3) = r(\beta_1, \beta_2, \beta_3) = 2$$

与 $r(\alpha_1, \alpha_2, \alpha_3) < r(\beta_1, \beta_2, \beta_3)$ 相矛盾. (舍)

$$\therefore a = 1. [\text{微信公众号: 考研斯基}]$$

注: 该题与本章【例 12】相同, 仅题干有些许变动, 再次出现是为了讲新解法.

【例 19】向量组 (I) $\alpha_1 = (1, 1, a)^T, \alpha_2 = (1, a, 1)^T$ 和 (II) $\beta_1 = (a, 1, 1)^T, \beta_2 = (4, 1, -5)^T$ 等价, 则 $a = \underline{\hspace{2cm}}$.

解: $\forall a, \beta_1, \beta_2$ 线性无关 $\Rightarrow r(\text{II}) = 2$

向量组 (I) 与 (II) 等价 $\Leftrightarrow r(\text{I}) = r(\text{II}) = r(\text{I}, \text{II})$.

$$[\alpha_1, \alpha_2, \beta_1, \beta_2] = \begin{bmatrix} 1 & 1 & a & 4 \\ 1 & a & 1 & 1 \\ a & 1 & 1 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & a & 4 \\ 0 & a-1 & 1-a & -3 \\ 0 & 0 & a^2+a-2 & 4a+8 \end{bmatrix}$$

$$\begin{cases} a-1 \neq 0, \\ a^2+a-2=0, \\ 4a+8=0, \end{cases} \therefore a = -2.$$

【例 20】(I) $\alpha_1, \alpha_2, \dots, \alpha_r$, (II) $\beta_1, \beta_2, \dots, \beta_s$, (I) 可由 (II) 表出 ()

(A) (I) 无关, 则 $r \leq s$

(B) (I) 相关, 则 $r > s$

(C) (II) 无关, 则 $r \leq s$

(D) (II) 相关, 则 $r > s$

解: $\because$ (I) 可由 (II) 表出

$$r(\alpha_1, \alpha_2, \dots, \alpha_r) \leq r(\beta_1, \beta_2, \dots, \beta_s) \leq s$$

如 (I) 无关, 则 $r(I) = r$

$\therefore r \leq s$ A 正确.

四、矩阵的秩

1、矩阵的秩 $r(A)$

矩阵 A 中非 0 子式的最高阶数.

$r(A) = r$: A 中有 r 阶子式不为 0, 每个 $r+1$ 阶子式 (如果有) 全为 0.

【定理 10】经初等变换矩阵的秩不变.

【定理 11】 $r(A) = A$ 的列秩 = A 的行秩.

【定理 12】 $Ax = 0$ 有非 0 解 $\Leftrightarrow r(A) < n$.

【定理 13】无关解向量的个数: $n - r(A)$. [微信公众号: 考研斯基]

求矩阵的秩的方法: ①行列式; ②相关或无关; ③方程组的解; ④特征值.

2、矩阵秩的公式

$$0 \leq r(A) \leq \min\{m, n\}$$

$$(1) r(A) = r(A^T) = r(AA^T) = r(A^T A)$$

$$(2) r(kA) = r(A), k \neq 0; r(A) = r(-A); r(A - E) = r(E - A)$$

$$(3) r(A + B) \leq r(A, B) \leq r(A) + r(B)$$

证明: 因 $(A, B) \begin{bmatrix} E \\ E \end{bmatrix} = A + B$

$$r(A + B) = r\left((A, B) \begin{bmatrix} E \\ E \end{bmatrix}\right) \leq r(A, B)$$

$$\text{又 } (E, E) \begin{bmatrix} A & O \\ O & B \end{bmatrix} = (A, B)$$

$$\therefore r(A, B) \leq r \begin{bmatrix} A & O \\ O & B \end{bmatrix} = r(A) + r(B)$$

$$(4) A - m \times n, B - n \times s$$

$$r(A) + r(B) - n \leq r(AB) \leq \min\{r(A), r(B)\}$$

$$(5) \text{若 } A \text{ 可逆}, r(AB) = r(B), r(BA) = r(B)$$

$$(6) \text{若 } A \text{ 列满秩, 则 } r(AB) = r(B)$$

$$\text{若 } B \text{ 行满秩, 则 } r(AB) = r(A)$$

$$(7) \text{若 } A - m \times n, B - n \times s, \text{ 且 } AB = 0, \text{ 则 } r(A) + r(B) \leq n$$

$$(8) A - m \times n, B - n \times s, C - s \times t, \text{ 则 } r(AB) + r(BC) \leq r(ABC) + n.$$

$$(9) \ r \begin{bmatrix} A & O \\ O & B \end{bmatrix} = r \begin{bmatrix} O & A \\ B & O \end{bmatrix} = r(A) + r(B)$$

$$r \begin{bmatrix} A & O \\ C & B \end{bmatrix} \geq r(A) + r(B)$$

证: $r \begin{bmatrix} A & B \\ O & B \end{bmatrix} = r(A) + r(B)$. [微信公众号: 考研斯基]

证明: 方法一 初等变换

$$\begin{bmatrix} A & B & | & E & O \\ O & B & | & O & E \end{bmatrix} = \begin{bmatrix} A & O & | & E & -E \\ O & B & | & O & E \end{bmatrix}$$

$$r \begin{bmatrix} A & B \\ O & B \end{bmatrix} = r \begin{bmatrix} A & O \\ O & B \end{bmatrix} = r(A) + r(B)$$

方法二 $\begin{bmatrix} E & -E \\ O & E \end{bmatrix} \begin{bmatrix} A & B \\ O & B \end{bmatrix} = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$

由 $\begin{bmatrix} E & -E \\ O & E \end{bmatrix}$ 可逆, 得 $r \begin{bmatrix} A & B \\ O & B \end{bmatrix} = r \begin{bmatrix} A & O \\ O & B \end{bmatrix} = r(A) + r(B)$

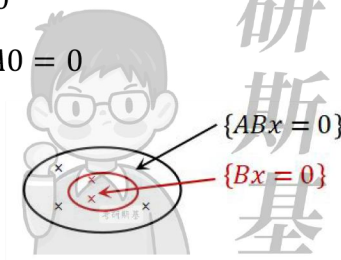
(10) 若 $A \sim B$, 则 $r(A) = r(B)$, $r(A + kE) = r(B + kE)$.

【公式证明】 $A - m \times n$, $B - n \times s$, 证明 $r(AB) \leq \min\{r(A), r(B)\}$.

证: 构造

$$(I) \ ABx = 0 \quad (II) \ Bx = 0$$

若 $B\alpha = 0 \Rightarrow AB\alpha = A(B\alpha) = A0 = 0$



$$s - r(B) \leq s - r(AB)$$

$$\therefore r(AB) \leq r(B)$$

$$\text{又 } r(AB) = r[(AB)^T] = r[B^T A^T] \leq r(A^T) = r(A)$$

在此基础上证: 若 A 可逆, $r(AB) = r(B)$

$$\text{若 } A \text{ 可逆, } r(AB) \leq r(B)$$

$$= r(EB)$$

$$= r[A^{-1}(AB)]$$

$$\leq r(AB)$$

$$\therefore r(AB) = r(B).$$

【补充】

$$\text{若 } x \leq y \leq z \leq t \leq x \Rightarrow x = y = z = t.$$

【例 21】 $A = \alpha\alpha^T + \beta\beta^T$, α, β 均 3 维列向量.

(1) 证 $r(A) \leq 2$

(2) 如 α, β 相关, 则 $r(A) < 2$.

证明: (1) $\alpha\alpha^T, \beta\beta^T$ 均 3 阶矩阵

且 $r(\alpha\alpha^T) \leq r(\alpha) \leq 1, r(\beta\beta^T) \leq 1$

$\therefore r(A) = r(\alpha\alpha^T + \beta\beta^T) \leq r(\alpha\alpha^T) + r(\beta\beta^T) \leq 2$.

(2) 如 α, β 相关, $\exists$ 不全为 0 的 k_1, k_2 使 $k_1\alpha + k_2\beta = 0$

不妨设 $k_2 \neq 0$, 则 $\beta = k\alpha$,

$r(A) = r[\alpha\alpha^T + (k\alpha)(k\alpha)^T] = r[(1 + k^2)\alpha\alpha^T] = r(\alpha\alpha^T) \leq 1 < 2$.

【例 22】 $A, B - n$ 阶, 则 ().

(A) $r(A, AB) = r(A)$

(B) $r(A, BA) = r(A)$

(C) $r(A, B) = \max\{r(A), r(B)\}$

(D) $r(A, B) = r(A^T, B^T)$

解: 如 A, B 均可逆, 4 个命题全正确, 故 A, B 不均可逆.

若 $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, r(A, B) = r \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = 2, \max\{r(A), r(B)\} = 1$

(C) 不正确.

$r(A^T, B^T) = r \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = 1$, (D) 不正确. [微信公众号: 考研斯基]

若 $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

$r(A, BA) = r \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = 2 \neq r(A)$, (B) 不正确.

【例 23】 $A, B, C - n$ 阶, $ABC = 0, \begin{bmatrix} O & A \\ BC & E \end{bmatrix}, \begin{bmatrix} AB & C \\ O & E \end{bmatrix}, \begin{bmatrix} E & AB \\ AB & O \end{bmatrix}$ 三个矩阵的秩分别为 r_1, r_2, r_3 , 判断 r_1, r_2, r_3 大小 ().

(A) $r_1 \leq r_2 \leq r_3$

(B) $r_1 \leq r_3 \leq r_2$

(C) $r_3 \leq r_1 \leq r_2$

(D) $r_2 \leq r_1 \leq r_3$

解: $\begin{bmatrix} O & A \\ BC & E \end{bmatrix} \rightarrow \begin{bmatrix} -ABC & O \\ BC & E \end{bmatrix} \rightarrow \begin{bmatrix} O & O \\ BC & E \end{bmatrix} \rightarrow \begin{bmatrix} O & O \\ O & E \end{bmatrix}, r_1 = n$.

$\begin{bmatrix} AB & C \\ O & E \end{bmatrix} \rightarrow \begin{bmatrix} AB & O \\ O & E \end{bmatrix}, r_2 = n + r(AB)$

$\begin{bmatrix} E & AB \\ AB & O \end{bmatrix} \rightarrow \begin{bmatrix} E & AB \\ O & -ABAB \end{bmatrix} \rightarrow \begin{bmatrix} E & O \\ O & -ABAB \end{bmatrix}, r_3 = n + r(ABAB)$

$r(AB) \leq \min\{r(A), r(B)\}$ 矩阵的秩越乘越小

$$\therefore r(ABAB) \leq r(AB)$$

$$\therefore r_1 \leq r_3 \leq r_2$$



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